



RICE

# NFL Coaching Value

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# Introduction

How do coaches add value, and why do we care?

**“Good players can't  
overcome bad  
coaching.**

— Bill Belichick





# Why do we care?

- Vrabel (Patriots) AFC Champion in year 1 (first since 2018).
- Macdonald (Seahawks) Superbowl winner in year 2 (first since 2014).
- Payton (Broncos) made AFCCG in year 3 (first since 2015).
- Super busy hiring cycle this offseason.
- No existing metric for coaches.

# How do coaches add value?

- **Player Development;** players exceeding expectations
- **Fourth Downs;** good decisions resulting in added epa/wp
- **Close Games;** good decisions resulting in added epa/wp
- **Regular Season / Playoffs;** raw success in regular and post-season
- **Divisional Games;** ability to beat familiar opponents
- **Clock Management;** good decisions resulting in added epa/wp

# Data & Methodology

# Data

⇒ PFR - Coaches (primary) by team-season ranging back for the entire super-bowl era.

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⇒ nflfastR - Play by play, season data, etc.

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⇒ nflreadr - Play by play, season data, etc.

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# Methodology

- ⇒ Create a series of independent “scores” tracking various coaching statistics/metrics.
- ⇒ Combine these scores into a larger cumulative score.
- ⇒ Use this data to highlight top coaching candidates for this year’s cycle.
- ⇒ Create a shiny application to allow users to specify score weights and features.



# Modeling

# Modeling Process

- **Close Games;** wp% over expected added on plays in 1-score games; percentile by season
- **Fourth Downs;** wp% over expected added on 4<sup>th</sup> down plays; percentile by season
- **Crunch Time;** wp% over expected added on plays with  $\leq 2$ min left; percentile by season
- **Regular Season;** percentile ranking of team's record for that season
- **Divisional Games;** percentile ranking of team's division game record for that season
- **Playoffs;** percentile ranking of team's playoff performance (based on what round they exited, if they made it)

# Player Development

## Hierarchical Bayesian APM

- Quantify how well a coach makes players “overperform” their expected value
- APM (Adjusted Plus-Minus) based on EPA
- Need to eliminate noise from teammates and game situations to find player’s true impact.
- Hierarchical Bayesian model treating every play as an observation of 22 players.
- Backups with small sample sizes aren’t over-evaluated, while starters with high snap counts provide precise data.
- Accounts for offense or defense and the baseline EPA of the situation.

# Player Development

## Defining “development”

- Development = Observed APM – Expected APM
- Expected APM ←
  - Regression based on prior performance, age and experience, draft position, snap counts
- Systemic overperformance across a roster is attributed to coaching quality

# Results & Discussion



# Top Seasons Since 2017

| rank | Team                                                                                      | Season | Score | Head Coach            | OC              | DC              |
|------|-------------------------------------------------------------------------------------------|--------|-------|-----------------------|-----------------|-----------------|
| 1    |  SF    | 2019   | 8.17  | <b>Kyle Shanahan</b>  |                 | Robert Saleh    |
| 2    |  DET   | 2024   | 8.14  | <b>Dan Campbell</b>   | Ben Johnson     | Aaron Glenn     |
| 3    |  PHI   | 2022   | 8.06  | <b>Nick Sirianni</b>  | Shane Steichen  | Jonathan Gannon |
| 4    |  NO    | 2018   | 7.99  | <b>Sean Payton</b>    | Pete Carmichael | Dennis Allen    |
| 5    |  BAL   | 2019   | 7.98  | <b>John Harbaugh</b>  | Greg Roman      | Don Martindale  |
| 6    |  LAR   | 2018   | 7.96  | <b>Sean McVay</b>     |                 | Wade Phillips   |
| 7    |  PHI | 2017   | 7.92  | <b>Doug Pederson</b>  | Frank Reich     | Jim Schwartz    |
| 8    |  KC  | 2020   | 7.84  | <b>Andy Reid</b>      | Eric Bieniemy   | Steve Spagnuolo |
| 9    |  PHI | 2024   | 7.83  | <b>Nick Sirianni</b>  | Kellen Moore    | Vic Fangio      |
| 10   |  BUF | 2024   | 7.80  | <b>Sean McDermott</b> | Joe Brady       | Bobby Babich    |

# Top Coaches

## Since 2017

| Coach                 | Seasons | Score | Teams                                                                                                                                                                                                                                                             |
|-----------------------|---------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Andy Reid</b>      | 9       | 59.97 |                                                                                                                                                                                |
| <b>Sean McDermott</b> | 9       | 53.68 |                                                                                                                                                                                |
| <b>Sean McVay</b>     | 9       | 49.52 |                                                                                                                                                                                |
| <b>Sean Payton</b>    | 8       | 48.60 |                                                                                             |
| <b>John Harbaugh</b>  | 9       | 47.79 |                                                                                                                                                                                |
| <b>Mike Tomlin</b>    | 9       | 47.21 |                                                                                                                                                                                |
| <b>Matt LaFleur</b>   | 9       | 43.99 |    |
| <b>Kyle Shanahan</b>  | 9       | 43.51 |                                                                                                                                                                              |
| <b>Nick Sirianni</b>  | 8       | 37.51 |                                                                                         |
| <b>Pete Carroll</b>   | 8       | 35.56 |                                                                                         |

# Top Coaches In 2025

Despite a strong season, McDermott was still fired. This is likely due to the Bill's inability to make a Superbowl despite their consistent success.

| rank |                                                                                       | Team  | Season | Score | Head Coach     | OC             | DC                |
|------|---------------------------------------------------------------------------------------|-------|--------|-------|----------------|----------------|-------------------|
| 1    |    | NE    | 2025   | 7.75  | Mike Vrabel    | Josh McDaniels | Terrell Williams  |
| 2    |    | BUF ★ | 2025   | 7.10  | Sean McDermott | Joe Brady      | Bobby Babich      |
| 3    |    | SEA   | 2025   | 6.84  | Mike Macdonald | Klint Kubiak   | Aden Durde        |
| 4    |    | HOU   | 2025   | 6.80  | DeMeco Ryans   | Nick Caley     | Matt Burke        |
| 5    |    | DEN   | 2025   | 6.70  | Sean Payton    | Joe Lombardi   | Vance Joseph      |
| 6    |    | LAR   | 2025   | 6.51  | Sean McVay     | Mike LaFleur   | Chris Shula       |
| 7    |  | JAX   | 2025   | 6.37  | Liam Coen      | Grant Udinski  | Anthony Campanile |
| 8    |  | LAC   | 2025   | 6.00  | Jim Harbaugh   | Greg Roman     | Jesse Minter      |
| 9    |  | CHI   | 2025   | 5.96  | Ben Johnson    | Declan Doyle   | Dennis Allen      |
| 10   |  | SF    | 2025   | 5.69  | Kyle Shanahan  | Klay Kubiak    | Robert Saleh      |

# Bottom Coaches

## In 2025

Other teams who parted from their coaches include:  
BUF (2; Sean McDermott, fired),  
PIT (11, Mike Tomlin, Resigned),  
ATL (18; Raheem Morris, fired),  
BAL (22; John Harbaugh, fired)

| rank | Team                                                                                                                                                                            | Season | Score | Head Coach             | OC                     | DC                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|------------------------|------------------------|-----------------------------|
| 23   |  CIN                                                                                         | 2025   | 2.96  | <b>Zac Taylor</b>      | Dan Pitcher            | Al Golden                   |
| 24   |  MIA      | 2025   | 2.94  | <b>Mike McDaniel</b>   | Frank Smith            | Anthony Weaver              |
| 25   |  CLE      | 2025   | 2.50  | <b>Kevin Stefanski</b> | Tommy Rees             | Jim Schwartz                |
| 26   |  NO                                                                                          | 2025   | 2.16  | <b>Kellen Moore</b>    | Doug Nussmeier         | Brandon Staley              |
| 27   |  TEN      | 2025   | 1.88  | <b>Brian Callahan</b>  | Nick Holz              | Dennard Wilson              |
| 28   |  NYG      | 2025   | 1.80  | <b>Brian Daboll</b>    | Mike Kafka             | Shane Bowen, Charlie Bullen |
| 29   |  WSH                                                                                       | 2025   | 1.69  | <b>Dan Quinn</b>       | Kliff Kingsbury        | Joe Whitt                   |
| 30   |  ARI  | 2025   | 1.25  | <b>Jonathan Gannon</b> | Drew Petzing           | Nick Rallis                 |
| 31   |  LV   | 2025   | 0.97  | <b>Pete Carroll</b>    | Chip Kelly, Greg Olson | Patrick Graham              |
| 32   |  NYJ                                                                                       | 2025   | 0.79  | <b>Aaron Glenn</b>     | Tanner Engstrand       | Steve Wilks                 |

# Hired HC This Cycle

- Bills – Joe Brady (2.53\*)
- Cardinals – Mike LaFleur (2.12\*)
- Falcons – Kevin Stefanski (3.46)
- Ravens – Jesse Minter (2.95\*)
- Browns – Todd Monken (2.011\*)
- Raiders – Klint Kubiak (2.28\*)
- Dolphins – Jeff Hafley (2.5\*)
- Giants – John Harbaugh (5.31)
- Steelers – Mike McCarthy (4.94)
- Titans – Robert Saleh (2.14)

\*only ever an asst coach, so score is out of 4.5



# Notable former HC (not hired)

- Sean McDermott (5.96)
- Doug Pederson (4.59)
- Pete Carroll (4.45)
- Mike McDaniel (4.29)
- Mike Zimmer (4.24)



# Notable Asst Coaches (not hired)

- Eric Bieniemy; 3.14
- Greg Roman; 2.9
- Pete Carmichael; 2.78
- George Edwards; 2.8
- Joe Lombardi; 2.51

# Future Work & Areas for Improvement

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- ⇒ More robust dataset of Coaches allows for more depth
- ⇒ Coaching network would be a great addition – lack of data depth prevents this currently
- ⇒ Inherently best when used with custom weights - Shiny App will improve this

# Conclusion

By using various EPA and wp% metrics, we can attribute these residuals to coaches to give an estimate of the literal value they add to their teams.

This information can then be used to not only identify top performing current coaches, but also up and coming assistant coaches that should be targets during a hiring cycle.

# Thank You!

Any questions?